# Lab Task – 1

## Reinforcement Learning Laboratory

### Experiment 1 : Exploring Reinforcement Learning Environments using Gymnasium

# Task 1 : Environment Setup

- Install Gymnasium, NumPy, and Matplotlib.
- Import the required libraries.
- Verify the installation by printing the Gymnasium version.

**Expected Output:** Successful installation and Gymnasium version displayed.

```
# Install required packages (run once; comment out after first successful run)
!pip install gymnasium numpy matplotlib -q
```

```
# Import required libraries
import gymnasium as gym
import numpy as np
import matplotlib.pyplot as plt

# Verify installation by printing the Gymnasium version
print(f"Gymnasium version: {gym.__version__}")
print(f"NumPy version: {np.__version__}")
```

```
Gymnasium version: 1.3.0
NumPy version: 2.0.2
```

# Task 2 : Create and Initialize an RL Environment

- Create the `CartPole-v1` environment.
- Reset the environment.
- Display the initial observation and environment information.

```
# Create the CartPole-v1 environment
env = gym.make("CartPole-v1")

# Reset the environment to obtain the initial state
initial_observation, info = env.reset(seed=42)

print("Environment created: CartPole-v1")
print(f"Initial Observation : {initial_observation}")
print(f"Info                : {info}")
```

```
Environment created: CartPole-v1
Initial Observation : [ 0.0273956  -0.00611216  0.03585979  0.0197368 ]
Info                : {}
```

# Task 3 : Explore Observation and Action Spaces

- Display the observation space.
- Display the action space.
- Display the type of observation space.
- Display the number of possible actions.

```
# Observation space
print("Observation Space           :", env.observation_space)
print("Observation Space Type      :", type(env.observation_space))
print("Observation Space Shape     :", env.observation_space.shape)
print("Observation Space Low       :", env.observation_space.low)
print("Observation Space High      :", env.observation_space.high)

print()

# Action space
```

```
print("Action Space                 :", env.action_space)
print("Number of Possible Actions  :", env.action_space.n)
```

```
Observation Space          : Box([-4.8               -inf -0.41887903        -inf], [4.8               inf 0.41887903
Observation Space Type     : <class 'gymnasium.spaces.box.Box'>
Observation Space Shape    : (4,)
Observation Space Low      : [-4.8               -inf -0.41887903        -inf]
Observation Space High     : [4.8               inf 0.41887903        inf]

Action Space               : Discrete(2)
Number of Possible Actions  : 2
```

**Interpretation (CartPole-v1):**

| Index | Observation | Min | Max |
|---|---|---|---|
| 0 | Cart Position | -4.8 | 4.8 |
| 1 | Cart Velocity | -Inf | Inf |
| 2 | Pole Angle (radians) | -0.418 | 0.418 |
| 3 | Pole Angular Velocity | -Inf | Inf |

The action space is discrete with 2 possible actions: `0` = push cart left, `1` = push cart right.

## Task 4 : Execute a Random Agent

- Execute a random action at every time step.
- Run the environment for one complete episode.
- Print step number, selected action, observation, reward, and episode status.
- Display the total number of steps and cumulative reward.

```
# Reset environment before starting a fresh episode
observation, info = env.reset(seed=42)

step_count = 0
cumulative_reward = 0.0
terminated = False
truncated = False

print(f"{'Step':<6}{'Action':<10}{'Observation':<55}{'Reward':<10}{'Terminated':<12}{'Truncated':<10}")
print("-" * 100)

while not (terminated or truncated):
    # Sample a random action from the action space
    action = env.action_space.sample()

    # Execute the action in the environment
    observation, reward, terminated, truncated, info = env.step(action)

    step_count += 1
    cumulative_reward += reward

    print(f"{step_count:<6}{action:<10}{str(np.round(observation, 4)):<55}"
          f"{reward:<10}{str(terminated):<12}{str(truncated):<10}")

print("-" * 100)
print(f"\nEpisode finished after {step_count} steps.")
print(f"Cumulative reward: {cumulative_reward}")

env.close()
```

```
Step  Action    Observation                                            Reward    Terminated  Truncated
----------------------------------------------------------------------------------------------------
1     1         [ 0.0273  0.1885  0.0363 -0.2614]                      1.0       False       False
2     0         [ 0.031  -0.0071  0.031   0.0425]                      1.0       False       False
3     1         [ 0.0309  0.1875  0.0319 -0.2403]                      1.0       False       False
4     0         [ 0.0347 -0.008   0.0271  0.0623]                      1.0       False       False
5     0         [ 0.0345 -0.2035  0.0283  0.3634]                      1.0       False       False
6     0         [ 0.0304 -0.3991  0.0356  0.6649]                      1.0       False       False
7     1         [ 0.0224 -0.2044  0.0489  0.3836]                      1.0       False       False
8     1         [ 0.0183 -0.01    0.0566  0.1067]                      1.0       False       False
9     1         [ 0.0181  0.1842  0.0587 -0.1676]                      1.0       False       False
10    0         [ 0.0218 -0.0117  0.0553  0.143 ]                      1.0       False       False
11    0         [ 0.0216 -0.2076  0.0582  0.4526]                      1.0       False       False
12    0         [ 0.0174 -0.4035  0.0673  0.7631]                      1.0       False       False
13    1         [ 0.0094 -0.2093  0.0825  0.4923]                      1.0       False       False
14    1         [ 0.0052 -0.0155  0.0924  0.2267]                      1.0       False       False
15    0         [ 0.0049 -0.2118  0.0969  0.547 ]                      1.0       False       False
16    1         [ 0.0006 -0.0181  0.1078  0.2864]                      1.0       False       False
17    1         [0.0003 0.1753 0.1136 0.0296]                          1.0       False       False
18    1         [ 0.0038  0.3686  0.1142 -0.2252]                      1.0       False       False
```

```
19    1         [ 0.0112  0.5619  0.1096 -0.4798]                      1.0       False       False
20    0         [ 0.0224  0.3655  0.1    -0.1547]                      1.0       False       False
21    0         [0.0297 0.1691 0.097  0.1678]                          1.0       False       False
22    1         [ 0.0331  0.3627  0.1003 -0.0928]                      1.0       False       False
23    1         [ 0.0403  0.5562  0.0985 -0.3522]                      1.0       False       False
24    0         [ 0.0515  0.3599  0.0914 -0.0302]                      1.0       False       False
25    0         [0.0587 0.1635 0.0908 0.2899]                          1.0       False       False
26    0         [ 0.0619 -0.0327  0.0966  0.6097]                      1.0       False       False
27    1         [0.0613 0.1609 0.1088 0.349 ]                          1.0       False       False
28    1         [0.0645 0.3543 0.1158 0.0925]                          1.0       False       False
29    1         [ 0.0716  0.5476  0.1176 -0.1615]                      1.0       False       False
30    1         [ 0.0825  0.7409  0.1144 -0.4149]                      1.0       False       False
31    0         [ 0.0974  0.5443  0.1061 -0.0885]                      1.0       False       False
32    0         [0.1082 0.3479 0.1043 0.2357]                          1.0       False       False
33    0         [0.1152 0.1514 0.109  0.5594]                          1.0       False       False
34    1         [0.1182 0.3448 0.1202 0.303 ]                          1.0       False       False
35    0         [0.1251 0.1482 0.1263 0.631 ]                          1.0       False       False
36    0         [ 0.1281 -0.0484  0.1389  0.9606]                      1.0       False       False
37    1         [0.1271 0.1446 0.1581 0.7146]                          1.0       False       False
38    0         [ 0.13   -0.0523  0.1724  1.0526]                      1.0       False       False
39    1         [0.129  0.1402 0.1935 0.8186]                          1.0       False       False
40    1         [0.1318 0.3322 0.2098 0.5925]                          1.0       True        False
----------------------------------------------------------------------------------------------------

Episode finished after 40 steps.
Cumulative reward: 40.0
```